

75%

ABERDARE TECHNICAL COLLEGE  
ICT TECHNICIAN LEVEL 5  
UNIT: COMPUTER NETWORKING AND DESIGN  
CONTINUOUS ASSESSMENT TEST (CAT) 3

TIME: 2 HOURS  
TOTAL MARKS: 50-65

NAME: MAIRU MARBLE

ADMISSION NO: \_\_\_\_\_

**INSTRUCTIONS**

1. Answer ALL questions.
2. Show all workings where applicable.
3. Write your answers clearly and neatly.

**QUESTION ONE (20 MARKS)**

1. Differentiate between the **OSI Model** and the **TCP/IP Model**. (2 Marks)
2. State any two functions of the **Network Layer** in the OSI model. (2 Marks)
3. Define the following terms: (2 Marks)
  - a) IPv4 Address
  - b) IPv6 Address
4. Identify the class of each of the following IP addresses: (2 Marks)
  - a) 10.0.0.1
  - b) 172.16.5.10
5. Differentiate between: (2 Marks)
  - a) Static IP Address
  - b) Dynamic IP Address
6. State any three advantages of IPv6 over IPv4. (3 Marks)
7. Explain the term **CIDR (Classless Inter-Domain Routing)**. (2 Marks)

15✓

8. State any two advantages of subnetting. (2 Marks)
9. Differentiate between a **Switch** and a **Router**. (3 Marks)

### QUESTION TWO (15 MARKS)

10. Outline three steps involved in the initial setup and configuration of a Cisco router. (3 Marks)
11. Write the Cisco IOS commands used to: (3 Marks)
- a) Enter global configuration mode.
  - b) Assign an IP address to an interface.
  - c) Save the running configuration.
12. Explain the purpose of the following routing protocols: (3 Marks)
- a) RIP
  - b) OSPF
  - c) EIGRP
13. State any two benefits of implementing VLANs in a network. (2 Marks)
14. What is an Access Control List (ACL) and why is it used in network security? (2 Marks)
15. Differentiate between NAT and PAT. (2 Marks)

### QUESTION THREE (15 MARKS)

16. State any two benefits of implementing Port Security on a switch. (2 Marks)
17. Explain the purpose of Spanning Tree Protocol (STP) in a switched network. (2 Marks)
18. What is Link Aggregation Control Protocol (LACP) and why is it used? (2 Marks)
19. Explain any three configurations performed when setting up a Wireless Access Point (AP). (3 Marks)
20. State the functions of the following network services: (2 Marks)
- a) DHCP
  - b) DNS
21. What is an SSID and why is it important in wireless networking? (2 Marks)
22. Describe two measures that can be used to secure a wireless network. (2 Marks)
23. Define the term **Network Privileges** and state one example. (2 Marks)

25

**QUESTION FOUR: WIRESHARK AND NETWORK SECURITY (10 MARKS)**

13

24. Using Wireshark packet captures, explain the **TCP Three-Way Handshake** by describing the following packets: (4 Marks)

- a) SYN
- b) SYN-ACK
- c) ACK
- d) State the purpose of the TCP Three-Way Handshake.

25. What is Wireshark and state two uses of Wireshark in network administration? (3 Marks)

26. A network administrator wants to display only TCP packets in Wireshark: (3 Marks)

- a) Write the display filter used. (1 Mark)
- b) State one advantage of packet analysis during network troubleshooting. (2 Marks)

**TOTAL: 50 MARKS**

**END OF PAPER**

13



1. The OSI model is a seven layer model that acts as a blueprint to the whole network while TCP/IP model is a four layer.
2. Network layer is responsible for logical addressing and Routing.
3. IPv4 is a traditional and most commonly used internet address. IPv6 is a newer version designed to solve the address shortage.
4. 10.0.0.1 Class A  
192.168.1.10 Class B
5. Static IP address is a fixed manually configured address that does not change while dynamic IP address is automatically assigned by DHCP server.
6. IPv6 has built-in security.  
It is simpler and more efficient.  
It has better auto configuration.
7. CIDR is a flexible way of representing IP addresses and their subnet masks using Prefix notation.

Saves IP address space by dividing large networks into smaller.  
Improves network security and performance by reducing broadcast.

Switch is a layer 2 and connects devices in same network.  
Router is a layer 3 and connects devices of different network.

Insert to console cable and open terminal.  
Enter on the router and enter setup mode.  
Configure hostname, interfaces and passwords.

19



11. Enter global configuration mode. - Configure terminal  
Assign IP address to interface. IP address 192.168.3.1 255.255.255.0  
Save running configuration. Copy running-config startup-config.

12. RIP - Distance-vector Protocol, uses hop count to find best Path  
OSPF - Link-State Protocol, uses Shortest Path algorithm for fast convergence.  
EIGRP - Hybrid Protocol, combines features of distance vector and link-state.

13. Improves network security by separating traffic.  
Reduces broadcast traffic and increases performance.

14. ACL It is used in network security to permit or deny traffic based on IP addresses and ports.

15. NAT - Translates Private IP address to Public ones.  
PAT - A NAT type that uses Port numbers to allow many devices share one Public IP address.

16. Prevents unauthorized devices from connecting to the switch.  
Enhances network security by limiting MAC address per port.

17. Purpose of STP Spanning Tree Protocol Prevents loops in Switched by blocking redundant paths.

An LACP is used to bundle multiple physical links into a logical link to increase bandwidth and provide redundancy.

network name. (SSID)

Security

IP address and channel.